

Mathematics refresher course

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March 2026

1 Units, Areas and Volumes

1. What is the perimeter and the area of a disk of radius r ?
2. What is the area and the volume of a ball of radius r ?
3. What is the area and the volume of a cylinder of radius r and height h ?
4. Change the units:
 - 0.4 m^3 into L.
 - 130 mm^2 into m^2 .
 - $2.7 \text{ m}^3/\text{h}$ into L/s .
5. What is the mass of a ball of radius 40cm full of water?
6. A tank has a cylindrical shape with a base of diameter 120cm and height 2m. A liquid is poured inside the tank at a flow rate 200 L/min. When will the tank be full?

2 Common functions

7. For each function among exp, log, sin and cos, answer the following questions.
 1. What is its domain of definition?
 2. Draw the graph of the function.
8. Complete the following identities.
 1. $e^{x+y} =$
 2. $e^{x-y} =$
 3. $\log(xy) =$
 4. $\log\left(\frac{x}{y}\right) =$
 5. $\log(x^n) =$
 6. $\sin^2(x) + \cos^2(x) =$

3 Equations

3.1 Quadratic equations

9. Solve for x : $x^2 + x = 6$.
10. Solve for y : $y^2 = 8y - 17$.
11. Solve for z : $\frac{2z^2 + 1}{z} = 3$

3.2 Equations of 2 unknowns

12. Solve the linear system for x and y :

$$\begin{cases} 3x + 4y &= 16 \\ 2x - 3y &= 22 \end{cases}$$

13. A rectangle has a perimeter of 13cm and an area of 10cm^2 . What are the lengths of its sides ?
14. Solve for x_1 and x_2 :

$$\begin{cases} 16x_1(1 - x_1) - 12x_1x_2 &= 0 \\ -4x_2 + 10x_1x_2 &= 0 \end{cases}$$

4 Derivatives

15. Compute the derivatives of the following functions.
 1. $f(x) = 4x + 7$
 2. $g(t) = -2t^2 + 9t + 1$
 3. $h(y) = ye^{-y}$
 4. $j(z) = \log(z^2 + 1)$
 5. $k(u) = \frac{\cos u}{\sin u}$
16. Show that $2\theta/\pi < \sin \theta$ for all $\theta \in (0, \pi/2)$
17. If we consider a quantity y with a given unit, what is the unit of $\frac{dy}{dt}$, where t is in s ? of $\frac{dy}{dx}$ where x is in m?

5 Integrals

18. Compute the following integrals.
 1. $\int_0^1 (x^2 + x + 1)dx$
 2. $\int_1^3 \frac{1+y^2}{y}dy$
 3. $\int_0^{\pi/2} \cos(t)dt$

4. $\int_0^{+\infty} \exp(-u) du$
5. $\int_1^x \log(v) dv$

6 Differential equations (ODE)

19. Solve the following differential equations.

1. $\frac{df}{dx} = 1$
2. $\frac{dx}{dt} = -2x + 1$
3. $mC_p \frac{dT}{dt} = hA(T_\infty - T) + P$
4. $\alpha \frac{dP}{dz} + \beta P + \gamma = 0$

20. For each of the ODE above, what is the solution with the following initial conditions ?

1. $f(0) = 1$
2. $x(0) = 10$
3. $T(0) = T_\infty$
4. $P(0) = 0$

7 Partial derivatives, vector calculus and PDEs

21. Compute the partial derivatives of the following functions. Deduce their gradients.

1. $f(x, y) = xy + x - y$
2. $g(x_1, x_2) = \frac{x_1 x_2}{1 + x_2}$
3. $h(t, x) = e^{-tx^2}$

22. Compute the divergence of the following vector fields.

1. $\vec{V}_1(x, y) = \begin{pmatrix} 2x + y \\ -3x - y \end{pmatrix}$
2. $\vec{V}_2(x, y) = \begin{pmatrix} -2xy \\ x^2 + y^2 \end{pmatrix}$

23. Solve the following PDE:

$$\begin{cases} \frac{\partial \rho}{\partial t} + v \frac{\partial \rho}{\partial x} = 0 \\ \rho(0, x) = \rho_0(x) \end{cases}$$

8 Matrices

24. Consider two matrices M and O both of dimension $n \times p$. Can the following expressions be computed? What is the resulting matrix dimension?

1. $M + O$
2. $M \times O$
3. $M \times O^T$
4. $M^T \times O$

25. Compute the product of:

$$\begin{pmatrix} 2 & 0 \\ -1 & -2 \\ 0 & 3 \end{pmatrix} \times \begin{pmatrix} 1 & 4 & -5 \\ 3 & -1 & 1 \end{pmatrix}$$

26. Compute the inverse of the matrix A :

$$A = \begin{pmatrix} 4 & 2 \\ -5 & -3 \end{pmatrix}$$

27. Write the system of equations from question 12 in matrix form to solve it.